

* Evaluación AM1: retos semanales

Presentación oral final

↳ Trabajo / grupos (20 min / grupo).

Propuesta de temas.

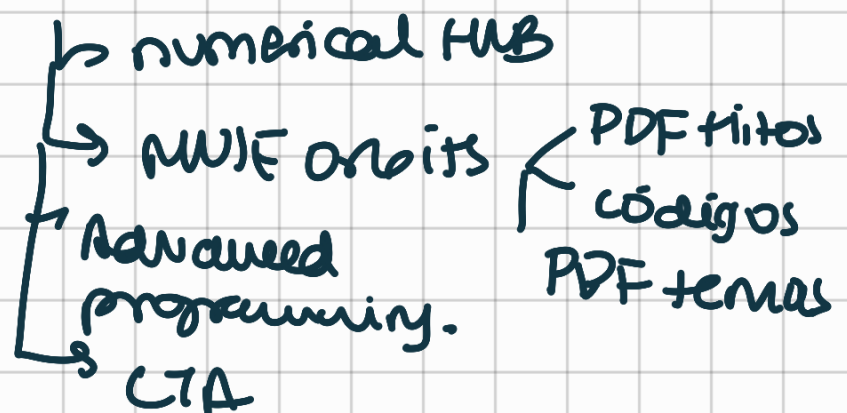
Trabajar paralelamente el
proyecto con los retos.

(10 - 15 min → info)

- Descifrando enigma.

Película: FIGURAS OCULTAS.
los piratas de silicon valley.
Micromen.

Bibliografía: gitHub - janr Work.



Visual studio code.

python

Julia

vincular
github.

clave acceso: j0u0b22

crear repositorio en github para
subir las colas.

visual studio - abrir carpeta en PC
que se puede
clonar en
un repositorio

1) Open OS

Revolution OS, The code
13th floor.

2) Nick Bostrom superinteligencia.

3) Social Dilemma

4) Her / Antonio Dionisio.

consciencia;

The triumph of nerds.

Advance programming (115) L

sonos una simulación - paper.

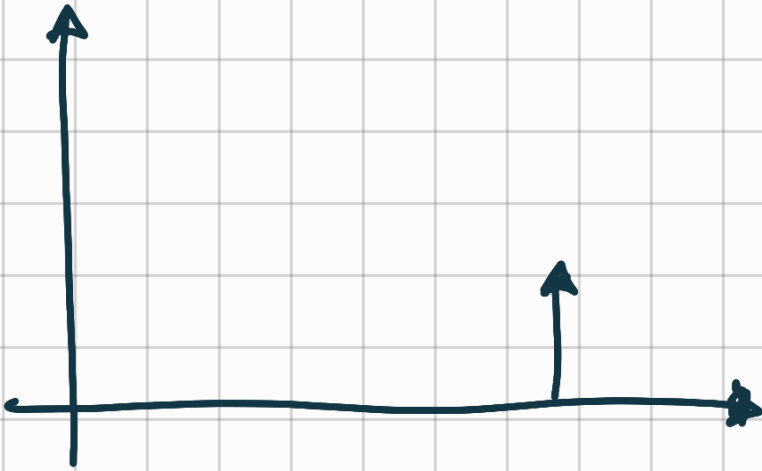
libro: por qué dormimos - Matthew Walter.

KEPLER ORBITS

$$\ddot{\vec{r}} = - \frac{\vec{r}}{\|\vec{r}\|^3}$$

$$\vec{r}(0) = (1, 0)$$

$$\dot{\vec{r}}(0) = (0, 1)$$



$$u = \begin{pmatrix} \vec{r} \\ \dot{\vec{r}} \end{pmatrix} \rightarrow \frac{du}{dt} = \begin{pmatrix} \dot{\vec{r}} \\ \ddot{\vec{r}} \end{pmatrix}$$

$$F = \frac{du}{dt} = \begin{pmatrix} \dot{\vec{r}} \\ -\frac{\vec{r}}{\|\vec{r}\|^3} \end{pmatrix} \quad (1)$$

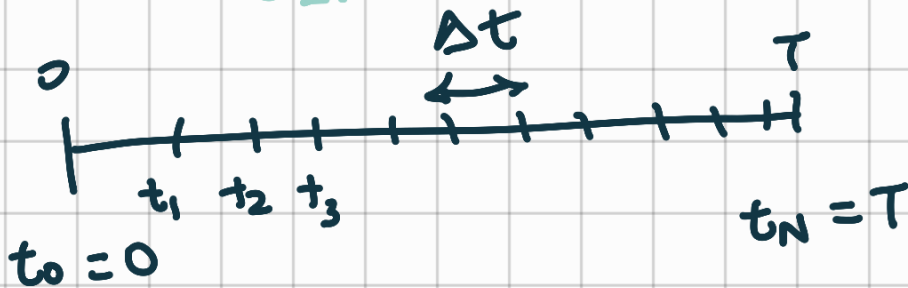
$\underbrace{\hspace{15em}}_{F(u,t)}$

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \dot{x} \\ \dot{y} \\ \ddot{x} \\ \ddot{y} \end{pmatrix} = \begin{pmatrix} \dot{x} \\ \dot{y} \\ -x/(x^2+y^2)^{3/2} \\ -y/(x^2+y^2)^{3/2} \end{pmatrix}$$

$$\boxed{\begin{aligned} \frac{du}{dt} &= F(u, t) \\ u(0) &= u^0 \end{aligned}}$$

C.I.

$$\int_{u^n}^{u^{n+1}} du = \int_{t_n}^{t_{n+1}} F(u, t) dt \quad *$$

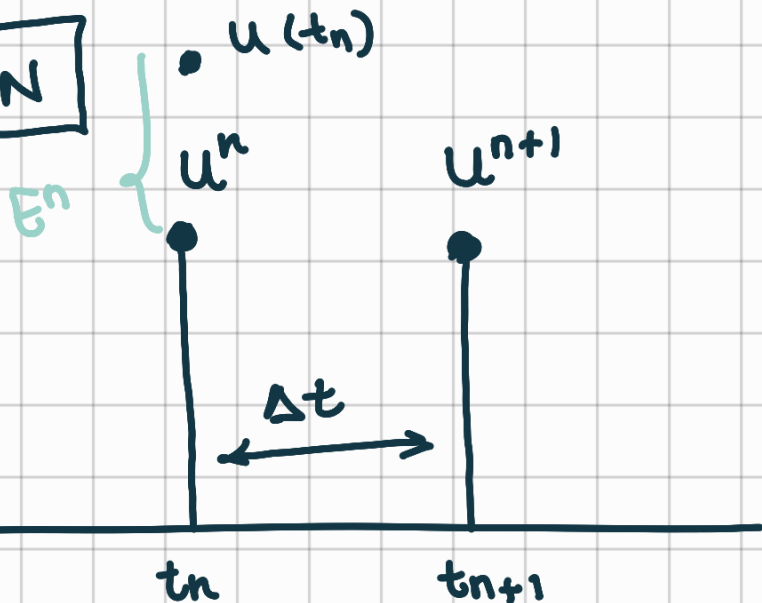


$N = n^\circ$ pasos

$n = \text{índice de pasos.}$

$$t_n = t_0 + \Delta t n$$

$$\boxed{T = \Delta t \cdot N}$$



$$\textcircled{*} \quad u^{n+1} - u^n \approx \underbrace{F(u^n, t_n)}_{F^n} (t_{n+1} - t_n)$$

$$\boxed{u^{n+1} = u^n + \Delta t F^n}; \quad n=1, \dots, N$$

* Hacer esta integración en Python:

1) Escalar: $u' = u^0 + \Delta t F^0 \quad (T=1, N=10)$

$$u^0 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Delta t = 0.1$$

$$u' = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} + 0.1 \cdot \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0.1 \\ -0.1 \\ 0 \end{pmatrix}$$

$$u' = \begin{pmatrix} 1 \\ 0.1 \\ -0.1 \\ 1 \end{pmatrix}$$

vector estado
en t_1

$$F' = \begin{pmatrix} -0.1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

$$u^2 = u' + \Delta t \cdot F'$$

=



* Escalar en python :

```
def F(x, y, xd, yd):
```

$$F_1 = x_d$$

$$F_2 = y_d$$

$$F_3 = -x / (x^2 + y^2)^{3/2}$$

$$F_4 = -y$$

```
return F1, F2, F3, F4
```

$$F_1, F_2, F_3, F_4 = F(x_n, y_n, x_{du}, y_{du})$$

$$x_{n1} = x_n + 0.1 \cdot F_1$$

$$y_{n1} = y_n + 0.1 \cdot F_2$$

$$x_{du1} = x_{du} + 0.1 \cdot F_3$$

$$y_{du1} = y_{du} + 0.1 \cdot F_4$$

2) Vectorial:

• numerical scheme: $\begin{cases} \text{implicit } G(u^{n+1}, \dots, u^{n-p+1}) = 0 \\ \text{explicit } \rightarrow u^{n+1} = G(u^n, u^{n-1}, u^{n-2}, \dots) \end{cases}$ Estabilidad?

implícito $\rightarrow$ cero-estable

coste computacional $\rightarrow$ cuántas operaciones se hacen en un determinado tiempo.

Explícito $\rightarrow u^{n+1} = u^n + \Delta t F(u^n, t_n)$

implícito $\rightarrow u^{n+1} = u^n + \Delta t F(u^{n+1}, t_{n+1})$

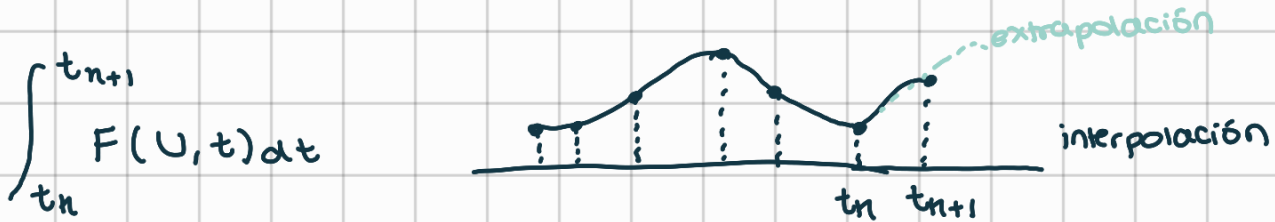
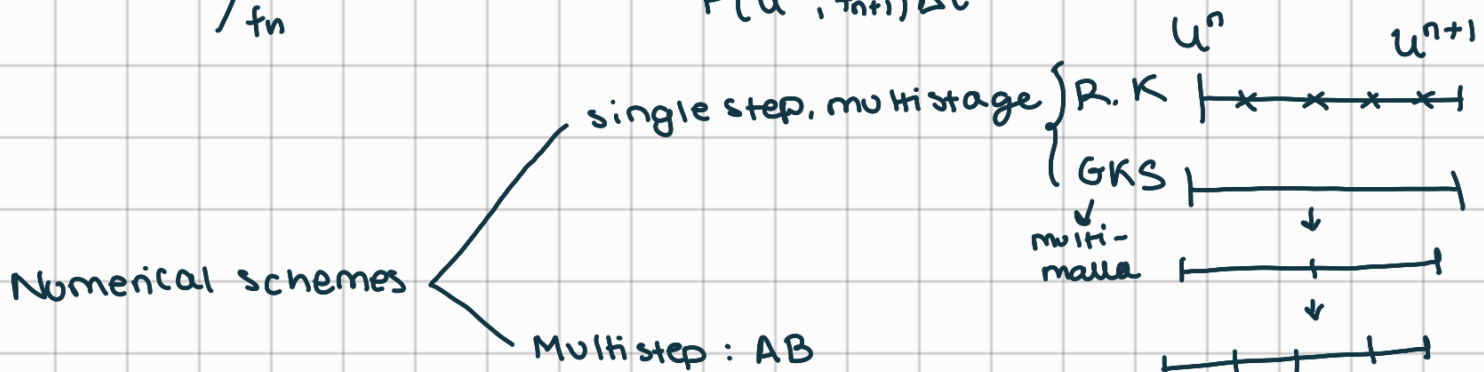
$$G = u^{n+1} - u^n - \Delta t F(u^{n+1}, t_{n+1})$$

$$G(x) = x - u^n - \Delta t F(x, t_{n+1})$$

$$\downarrow$$

$$x = u^{n+1}$$

$$\int_{t_n}^{t_{n+1}} F(u, t) dt \approx \begin{cases} F(u^n, t_n) \Delta t \\ F(u^{n+1}, t_{n+1}) \Delta t \end{cases}$$



* splines = interpolantes a trozos.

* round-off: error de redondeo.

(1) $\frac{du}{dt} = F(u, t)$

$u^{n+1} = G(u^n, u^{n-1}, u^{n-2}, \dots, u^{n-p+1})$

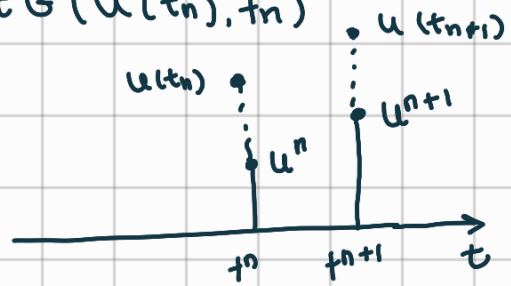
$$V^n = \begin{pmatrix} u^n \\ u^{n-1} \\ \vdots \\ u^{n-p+1} \end{pmatrix}$$

$u^n = G(V^{n-1})$

$V^{n+1} = H(V^n, t_n)$

$$(2) \quad U^{n+1} = U^n + \Delta t G(U^n, t_n)$$

$$(3) \quad T^{n+1} = U(t_{n+1}) - U(t_n) - \Delta t G(U(t_n), t_n)$$



$$(4) \quad E^{n+1} = U(t_{n+1}) - U^{n+1}$$

$$(2) \quad U^{n+1} - U^n - \Delta t G(U^n, t_n) = 0$$

$$(3) \quad U(t_{n+1}) - U(t_n) - \Delta t G(U(t_n), t_n) = T^{n+1}$$

Ej: Euler T^{n+1}

$$T^{n+1} = U(t_{n+1}) - U(t_n) - \Delta t F(U(t_n), t_n)$$

$$= U(t_n) + \left(\frac{dU}{dt} \right)_{t_n} \Delta t + \left(\frac{d^2U}{dt^2} \right)_{t_n} \frac{\Delta t^2}{2} + \dots - U(t_n) - \Delta t F(U(t_n), t_n)$$

$$(3) - (2) \quad \underbrace{U(t_{n+1}) - U^{n+1}}_{E^{n+1}} - \underbrace{[U(t_n) - U^n]}_{E^n} - \Delta t \left[\underbrace{G(U(t_n), t_n) - G(U^n, t_n)}_{A U(t_n) - A U^n} \right] = T^{n+1}$$

$$\frac{A[U(t_n) - U^n]}{E^n}$$

$$E^{n+1} = E^n + \Delta t A E^n + T^{n+1}$$

$$E^{n+1} = \underbrace{(I + \Delta t A)}_B E^n + T^{n+1}$$

$$E^{n+1} = B E^n + T^{n+1}$$

$$E^0 = U(t_0) - U^0$$

$$\|E^0\| = \varepsilon$$

$$E^1 = B E^0 + T^1$$

$$E^2 = B E^1 + T^2 = B(B E^0 + T^1 + T^2) = B^2 E^0 + B T^1 + T^2$$

$$E^3 = BE^2 + T^3 = B(B^0 E^0 + BT^1 + T^2) + T^3 = B^3 E^0 + B^2 T^1 + BT^2 + T^3$$

$$E^n = B^n E^0 + \sum_{k=1}^n T^k B^{n-k}$$

$$\|A\| = \sup_{\forall x} \frac{\|Ax\|}{\|x\|}$$

$$\|E^n\| \leq \|B^n E^0\| + \sum_{k=1}^n T^k B^{n-k}$$

$$\leq \|B^n\| \|E^0\| + \sum_{k=1}^n \|B^{n-k}\| \|T^k\| \leq \|B^n\| \|E^0\| + \sup_{\forall k} \|T^k\| \sum_{k=1}^n \|B^{n-k}\|$$

Si $B^T B = B B^T \Rightarrow \|B^n\| = \rho^n$
(normal)

→ RADIO ESPECTRAL
 $\rho = \sup |r|$

$$r = \lambda(B)$$

↓
n° máximo de los
autovalores $\Rightarrow \rho$.
↓
de B.

$$\|E^n\| \leq \rho^n E + \sup_{\forall k} \|T^k\| \sum_{k=1}^n \rho^{n-k}$$

$$S_n = 1 + \rho + \rho^2 + \rho^3 + \dots + \rho^{n-1}$$

$$\rho S_n = \rho + \rho^2 + \dots + \rho^n$$

$$(1-\rho) S_n = 1 - \rho^n$$

$$\|E^n\| \leq \rho^n E + \sup_{\forall k} \|T^k\| \left(\frac{1 - \rho^n}{1 - \rho} \right)$$

si $\rho > 1$: $n \rightarrow \infty \quad \|E^n\| \rightarrow \infty$

si $\rho < 1$: $n \rightarrow \infty \quad \|E^n\| \leq \frac{\sup \|T^k\|}{1 - \rho}$

si $\rho = 1$: $n \rightarrow \infty$

$$\Delta t = n \Delta t$$

$$\rho = 1 \quad \|E^n\| \leq E + \sup \|T^k\| n$$

$$\|E^n\| \leq E + \sup \|T^k\| \frac{\sigma}{\Delta t}$$

* esto es para entender, no para implementar.

* Planteamiento para conocer el error sin conocer la solución

$$\|E\|^n \leq \rho^n E + \sup \|T^K\| \left(\frac{1-\rho^n}{1-\rho} \right)$$

1) solución ϕ^0

2) solución aproximada $\phi(h)$

$\downarrow$
h: step.

RICHARDSON (1927)

(1) $\phi'(h_1) = \phi^0 + K_g h_1^q + O(h_1^{q+1})$

(2) $\phi^2(h_2) = \phi^0 + K_g h_2^q + O(h_2^{q+1})$

se restan

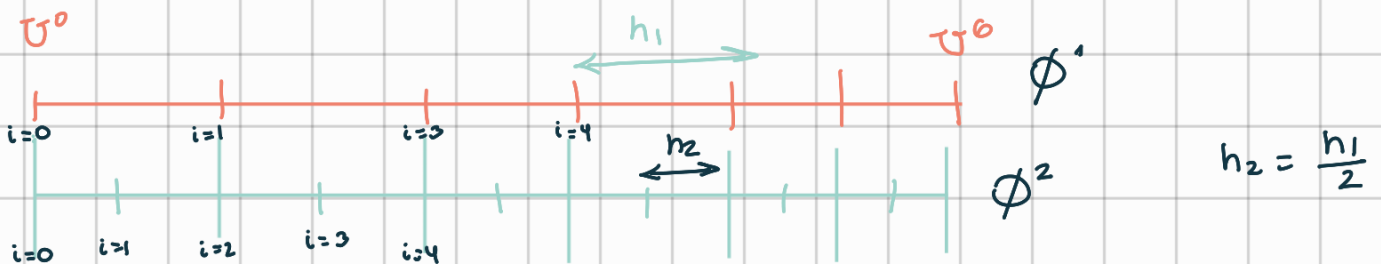
$$p(h) = \phi^0 + K_g \cdot h^q + O(h^{q+1})$$

$$h = 0.1; \quad h^2 = 10^{-2}; \quad h^3 = 10^{-3}$$

$$(1) - (2) \rightarrow \phi' - \phi^2 = K_g (h_1^q - h_2^q) + O(h_1^{q+1}, h_2^{q+1})$$

$$K_g \approx \left(\frac{\phi' - \phi^2}{h_1^q - h_2^q} \right)$$

$$E' = \phi^0 - \phi' = \left(\frac{\phi^2 - \phi'}{h_1^q - h_2^q} \right) h_1^q + O(h_1^{q+1})$$



$$\left(\frac{\phi' - \phi^2}{h_1^q - \left(\frac{h_1}{2}\right)^q} \right) h_1^q = \left(\frac{\phi' - \phi^2}{1 - \frac{1}{2^q}} \right) \rightarrow \text{medir el error en cada instante temporal de la malla gruesa}$$

$$\phi_i^1 - \phi_{2i}^2 \xrightarrow{\text{código}} U_1[i, :] - U_2[2 \times i, :]$$

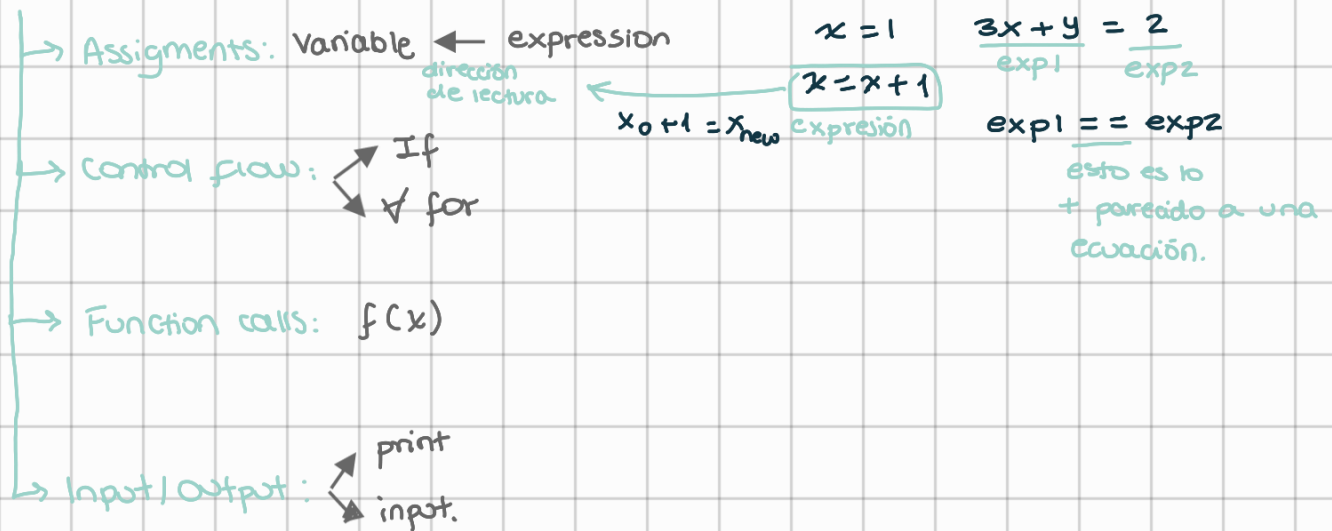
$$U[0:N, 1:Nv]$$

i $\swarrow \searrow$ índices vector de estados

filas $\rightarrow$ son instantes temporales

columnas $\rightarrow$ componentes del vector de estados en cada instante temporal.

* Statements (sentencias): Formado por 4 grupos.



Euler methods:

Euler method

$$\ddot{x} + x = 0$$

$$x(0) = 1; \dot{x}(0) = 0$$

$$U = \begin{pmatrix} x \\ \dot{x} \end{pmatrix} \quad \frac{d}{dt} \begin{pmatrix} x \\ \dot{x} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ \dot{x} \end{pmatrix}$$

$$\frac{dU}{dt} = F(U, t) = AU$$

$$A \cdot W = \lambda W$$

↓
autovalores

$$(A - \lambda I)W = 0$$

$$|A - \lambda I| = 0 \quad \leftarrow \text{AUTOVALORES.}$$

$$\begin{vmatrix} -\lambda & 1 \\ -1 & \lambda \end{vmatrix} = 0 \rightarrow \lambda^2 + 1 = 0 \quad \begin{cases} \lambda_1 = +i \\ \lambda_2 = -i \end{cases}$$

$$\begin{pmatrix} -i & 1 \\ -1 & -i \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow w_1 = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$\begin{pmatrix} +i & 1 \\ -1 & +i \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow w_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$U(t) = C_1 w_1 e^{\lambda_1 t} + C_2 w_2 e^{\lambda_2 t} \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ i \end{pmatrix} + C_2 \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$C_1 + C_2 = 1 \rightarrow C_2 = \frac{1}{2}$$

$$iC_1 - iC_2 = 0 \rightarrow iC_1 = iC_2 \rightarrow C_1 = C_2$$

$$U(t) = \frac{1}{2} \left[\begin{pmatrix} 1 \\ i \end{pmatrix} e^{it} + \begin{pmatrix} 1 \\ -i \end{pmatrix} e^{-it} \right] = \begin{pmatrix} \frac{1}{2}(e^{it} + e^{-it}) \\ \frac{1}{2}(e^{it} - e^{-it}) \end{pmatrix} = \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}$$

$$U^{n+1} = U^n + \Delta t F^n = U^n + \Delta t A U^n = \underbrace{(I + \Delta t A)}_B U^n \rightarrow U^{n+1} = B U^n \quad B = \begin{pmatrix} 1 & \Delta t \\ -\Delta t & 1 \end{pmatrix}$$

$$B W = r W$$

autovectores de B

$$|B - rI| = 0 \quad \begin{vmatrix} 1-r & \Delta t \\ -\Delta t & 1-r \end{vmatrix} = 0$$

$$(1-r)^2 + \Delta t^2 = 0 \quad \begin{cases} r_1 = 1 + \Delta t i \\ r_2 = 1 - \Delta t i \end{cases}$$

$$A W = \lambda W$$

¿Qué relación hay entre ambos?

$$(I + \Delta t A) W = r W$$

$$\Delta t A W = (r-1) W \rightarrow A W = \frac{r-1}{\Delta t} W$$

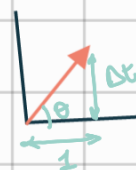
Teorema de transformación espectral

* De r_1 y r_2 no se van a sacar números, se van a obtener funciones.

$$A W = \lambda W \Rightarrow \lambda = \frac{r-1}{\Delta t}$$

$$r = 1 + \lambda \Delta t \quad \begin{cases} r_1 = 1 + \Delta t i \\ r_2 = 1 - \Delta t i \end{cases}$$

módulo - argumento



$$\begin{aligned} r_1 &= p e^{i\theta} \\ r_2 &= p e^{-i\theta} \\ p &= \sqrt{1 + \Delta t^2} \\ \tan \theta &= \Delta t \end{aligned}$$

$$\begin{aligned} U^n &= \kappa_1 W_1 r_1^n + \kappa_2 W_2 r_2^n \\ r_1^{n+1} W_1 &= r_1^n B W_1 \rightarrow B W = r W \\ U^n &= \kappa_1 \begin{pmatrix} 1 \\ i \end{pmatrix} \rho^n e^{in\theta} + \kappa_2 \begin{pmatrix} 1 \\ -i \end{pmatrix} \rho^n e^{-in\theta} \end{aligned}$$

$$U^0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \kappa_1 \begin{pmatrix} 1 \\ i \end{pmatrix} + \kappa_2 \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$\begin{aligned} \kappa_1 + \kappa_2 &= 1 \\ (\kappa_1 - \kappa_2)i &= 0 \end{aligned} \quad \left| \quad \kappa_1 = \kappa_2 = \frac{1}{2} \right.$$

$$U^n = \frac{\rho^n}{2} \begin{pmatrix} e^{in\theta} + e^{-in\theta} \\ ie^{in\theta} - ie^{-in\theta} \end{pmatrix}$$

$$= \frac{\rho^n}{2} \begin{pmatrix} 2 \cos(in\theta) \\ 2i \sin(in\theta) \end{pmatrix} = \begin{pmatrix} \rho^n \cos(in\theta) \\ -\rho^n \sin(in\theta) \end{pmatrix} = \begin{pmatrix} (1 + \Delta t^2)^{\frac{n}{2}} \cos(n\theta) \\ -(1 - \Delta t^2)^{\frac{n}{2}} \sin(n\theta) \end{pmatrix} \quad \begin{aligned} \tan \theta &= \Delta t \\ \theta &\approx \Delta t \end{aligned}$$

* Esquema numérico con error en amplitud

$$E^n = \alpha(t_n) - x^n = \cos(n\Delta t) - (1 + \Delta t^2)^{n/2} \cos(n\theta)$$

* Error de fase:

$$\tan \theta = \Delta t$$

$$\theta \approx \Delta t + O(\Delta t^3)$$

$$\theta - \frac{\theta^3}{3} \approx \Delta t$$

↓
pintándolo en Python
se obtienen bolitas
distanciadas entre
sí por el Δt definido

Teorema Transferencia Espectral

Sea $A \in M_{N \times N}$ y $q(x)$ un polinomio $\Rightarrow$ el espectro de $B = q(A)$ es $r = q(\lambda)$ $\lambda \in \Lambda(A)$

$$q(x) = \alpha_0 + \alpha_1 x + \dots + \alpha_n x^n$$

$$B = \alpha_0 + \alpha_1 A + \dots + \alpha_n A^n$$

$$P^{-1} B P = \alpha_0 P^{-1} P + \alpha_1 P^{-1} A P + \dots + \alpha_n P^{-1} A^n P$$

$$P^{-1} B P = \alpha_0 I + \alpha_1 \Lambda + \dots + \alpha_n \Lambda^n$$

A es diagonalizable

$$\exists P \mid P^{-1} A P = \Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

$$\Lambda^n = \begin{pmatrix} \lambda_1^n & & 0 \\ & \ddots & \\ 0 & & \lambda_n^n \end{pmatrix}$$

$$\begin{pmatrix} r_1 & & 0 \\ & \ddots & \\ 0 & & r_n \end{pmatrix} = P^{-1} B P = \alpha_0 I + \alpha_1 \Lambda + \dots + \alpha_n \Lambda^n = \alpha_0 \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} + \alpha_1 \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} + \dots + \alpha_n \begin{pmatrix} \lambda_1^n & & 0 \\ & \ddots & \\ 0 & & \lambda_n^n \end{pmatrix}$$

$$r_i = \alpha_0 + \alpha_1 \lambda_i + \dots + \alpha_n \lambda_i^n$$

Polinomio característico de estabilidad de un esquema

EJ $U^{n+1} = U^n + \Delta t F^n$

Euler

$$\Pi(r, w) = r - 1 - w$$

$$U^{n+1} - U^n = \Delta t F^n$$

$r - 1 - w$ ΔU^n

relación
axiomática.

EJ $U^{n+1} = U^n + \Delta t F^{n+1}$

Euler inverso

$$\Pi(r, w) = r - 1 - w r$$

EJ $U^{n+1} = U^n + \frac{\Delta t}{2} (F^n + F^{n+1})$

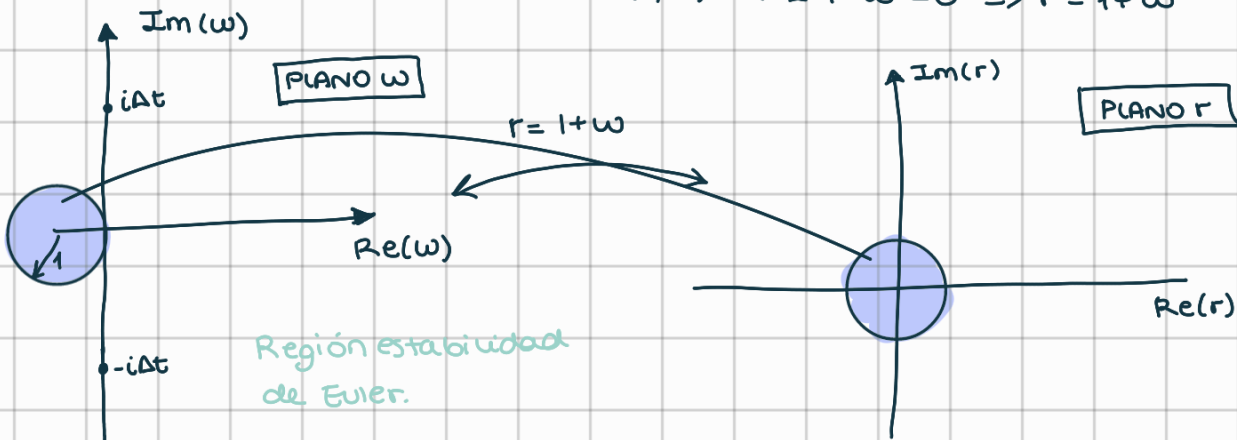
Nicholson

$$\Pi(r, w) = r - 1 - \frac{w}{2} (1 + r)$$

$$\mathcal{R}_A = \{ w \in \mathbb{C} \mid \Pi(r, w) = 0, |r_i| \leq 1 \}$$

Euler

$$\Pi(r, w) = r - 1 - w = 0 \Rightarrow r = 1 + w$$

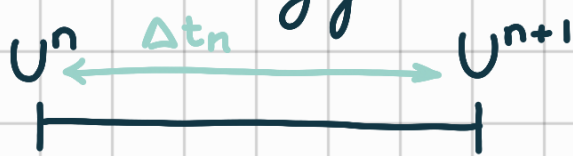


① si $\forall \lambda_i \in \mathcal{L}(A)$ $\lambda_i \Delta t \in \mathbb{R}_A$
 entonces $\mathcal{U}^n$ es asintóticamente estable

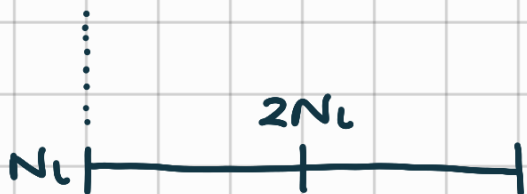
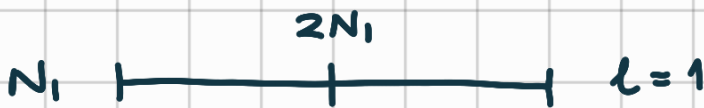
Oscilador $\ddot{x} + x = 0$
 $-\lambda_1 = +i$
 $\lambda_2 = -i$

② si $\exists \lambda_i \in \mathcal{L}(A)$
 $\lambda_i \Delta t \notin \mathbb{R}_A$
 entonces $\mathcal{U}^n$ es inestable.

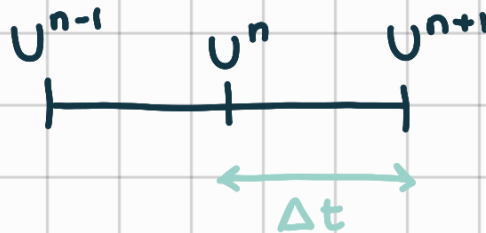
GBS: Gragg - Burlish - Steer)



$$h_L = \frac{\Delta t_n}{2N_L}$$



LEAP - FROG



Dato $\mathcal{U}^0$

$$\left\{ \begin{array}{l} U^1 = U^0 + \Delta t F^0 \text{ (EULER)} \\ U^2 = U^0 + 2\Delta t F^1 \\ U^3 = U^1 + 2\Delta t F^2 \end{array} \right.$$

$$U^{n+1} = U^{n-1} + 2\Delta t F^n; n=1, \dots$$

$$\left\{ \begin{array}{l} U^2 = U^0 + 2\Delta t F^1 \\ U^3 = U^1 + 2\Delta t F^2 \end{array} \right.$$

$$T^{n+1} = U(t_{n+1}) - U(t_{n-1}) - 2\Delta t F(U(t_n))$$

def Leap-Frog(F, t, U):

if $t_1 == 0$:

$$U_1 = U + (t_2 - t_1) * F$$

return U_1

else:

$$U_2 = U_1 + 2 * (t_2 - t_1) * F(U)$$

$U_1 = U$ * U_1 guardar como variable global $\rightarrow$ Que U_1 viva fuera de la función.

* Vector multipaso hacer vector de un paso con todos los pasos.

* Error local

$$T^{n+1} = U(t_{n+1}) - U(t_{n-1}) - 2\Delta t F(U(t_n))$$

(*)

$$= U(t_n) + \frac{dU}{dt}(t_n)\Delta t + \frac{d^2U}{dt^2}(t_n)\frac{\Delta t^2}{2} + \dots$$

$$- U(t_n) + \frac{dU}{dt}(t_n)\Delta t - \frac{d^2U}{dt^2}(t_n)\frac{\Delta t^2}{2} +$$

$$+ \frac{d^3U}{dt^3}(t_n)\frac{\Delta t^3}{6} + \dots$$

$$- 2\Delta t F(U(t_n))$$

$$= \frac{1}{3} \frac{d^3U}{dt^3}(t_n)\Delta t^3 + O(\Delta t^5)$$

$$(*) \quad U(t_{n-1}) = U(t_n - \Delta t) = U(t_n) + \frac{dU}{dt}(t_n)(-\Delta t)$$

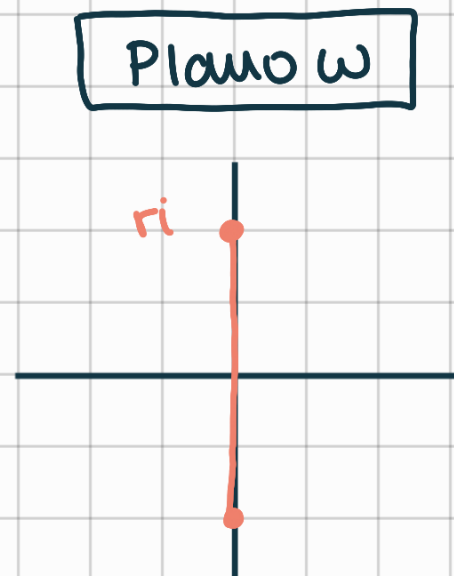
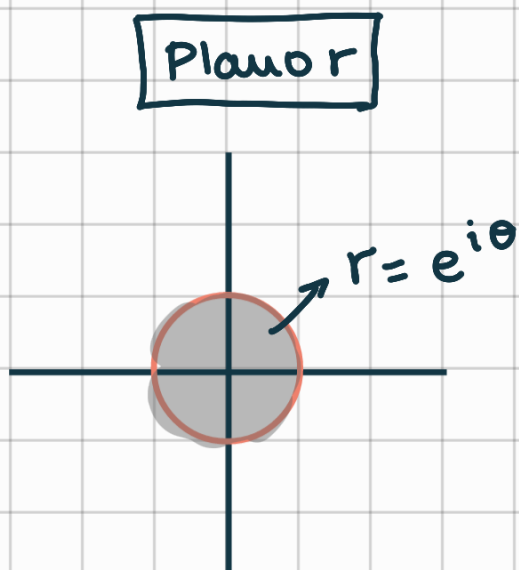
Estabilidad: $U^n = K r^n$

$$\Pi(r, \omega) = r^2 - 1 - 2\omega r$$

$$K r^{n+1} = K r^{n-1} + 2 \Delta t \lambda^{\omega} K r^n$$

$$r^2 = 1 + 2\omega r$$

$$R_A = \{ \omega \in \mathbb{C} \mid \Pi(r, \omega) = 0 \mid r| < 1 \}$$



$$r^2 - 1 - 2\omega r = 0$$

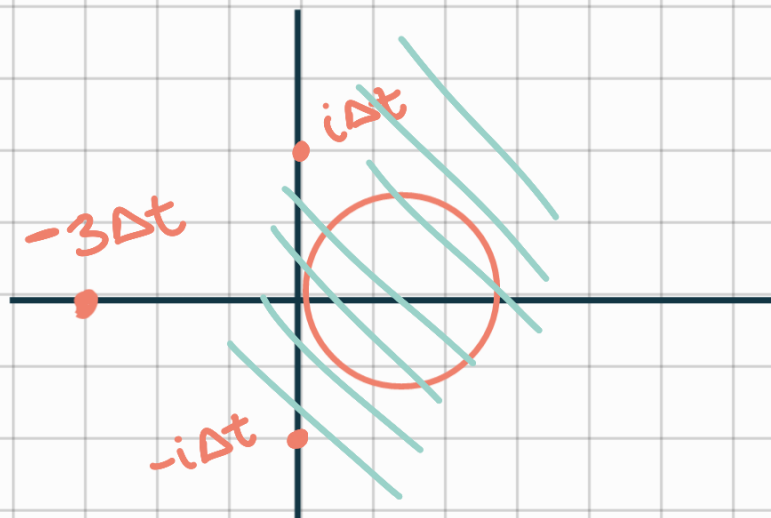
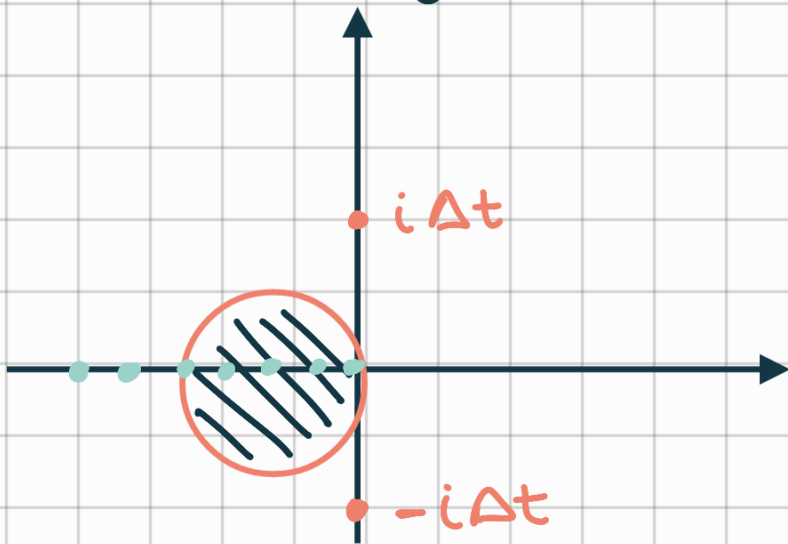
$$2\omega r = r^2 - 1$$

$$\omega = \frac{r^2 - 1}{2r} \stackrel{r=e^{i\theta}}{=} \frac{e^{2i\theta} - 1}{2e^{i\theta}} = \frac{1}{2}(e^{i\theta} - e^{-i\theta})$$

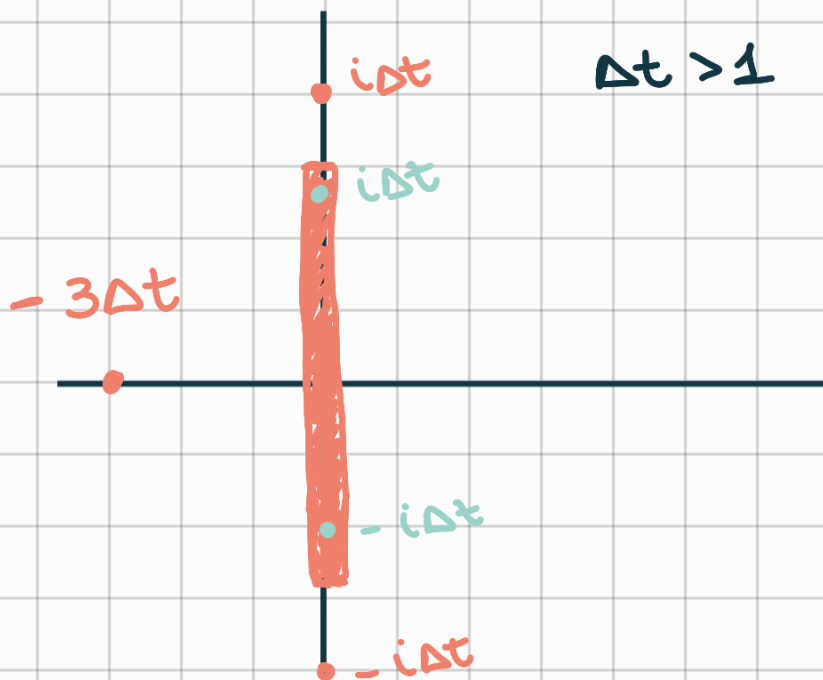
$$= \frac{2i \sin \theta}{2}$$

$$\frac{du}{dt} = -3u e^{-3t}$$

EULER

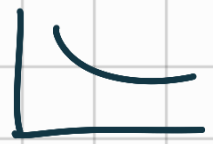


$\Delta t > 1$



RK buenos
son tangentes
al eje Im.

$$\ddot{x} + x = 0 \begin{cases} \lambda_1 = +1 \\ \lambda_2 = -1 \end{cases} \quad \omega = i\Delta t$$



$$U_0$$

$$U_1 = U_0 + h \epsilon F_0$$

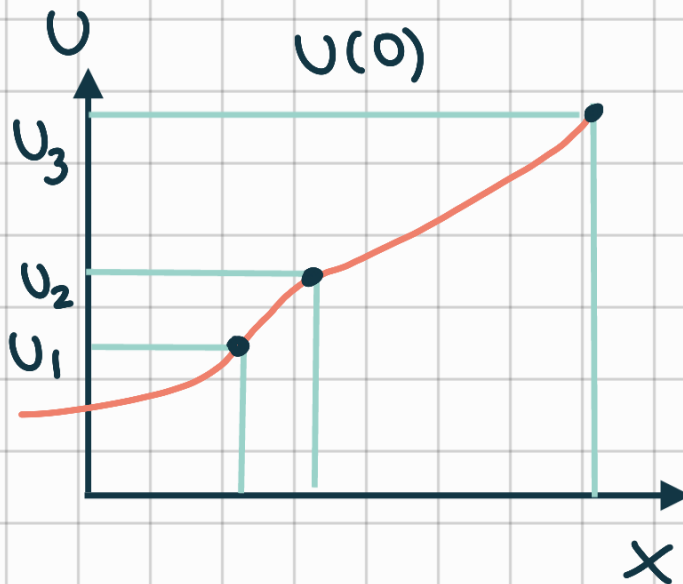
$$U_2 = U_0 + 2h \epsilon F_1$$

$$U_{2N_L+1} = U_{2N_L-1} + 2h \epsilon F_{2N_L}$$

$$U(h) = a_0 h^2 + a_1 h^2 + a_1 h^4 + \dots + a_g h^{2g}$$

$$h^2 = x$$

$$U(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_g x^g$$



$$U_q(h) = a_0 + a_1 h^2 + a_2 h^4 + \dots a_q h^{2q}$$

$$U_{q+1}(h) = a_0 + a_1 h^2 + \dots a_q h^{2q} + a_{q+1} h^{2q+2}$$

$$\|U_{q+1} - U_q\| < \varepsilon$$

si esto no se cumple se sigue bajando.

* GBS $\rightarrow$ no está en hitos

* Entrega $\rightarrow$

$$* z = y[y < 0.1]$$

* oscilador armónico

$$\left. \begin{array}{l} \ddot{x} + x = 0 \\ x(0) = 1; \dot{x}(0) = 0 \end{array} \right\} x(t) = \cos(t)$$

$$\lambda_1 = i$$

$$\lambda_2 = -i$$

* Espuña

* Meter esquema
(EAP FROG)

$$\frac{dz}{dt} = F(z, t)$$

↓
puede ser cualquier cosa

$$z_{n+1} = z_{n-1} + 2\Delta t F(z_n), \quad n = 1, 2, 3, \dots$$

$$z_0$$

$$z_1 = z_0 + \Delta t F(z_0)$$

$$\underbrace{\frac{d}{dt} \begin{pmatrix} x \\ \dot{x} \end{pmatrix}}_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ \dot{x} \end{pmatrix}$$

$$z = x + i\dot{x} ; \frac{dz}{dt} = \dot{x} + i\ddot{x} = \dot{x} - ix$$

$$= -i \underbrace{(x + i\dot{x})}_z$$

$$\frac{dz}{dt} = -iz$$

$$z_{n+1} = z_{n-1} - 2\Delta t i z_n ; n = 1, 2, \dots$$

$$z_n = C r^n \rightarrow z_n = C_1 r_1^n + C_2 r_2^n$$

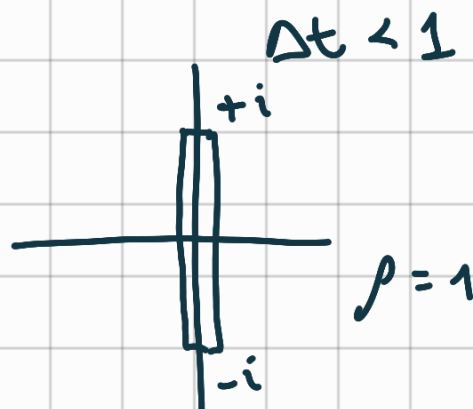
$$C r^{n+1} = C r^{n-1} - 2\Delta t i C r^n$$

$$r^2 = 1 - 2\Delta t i r$$

$$r^2 + 2\Delta t i r - 1 = 0$$

$$r_1 = -\Delta t i + \sqrt{-\Delta t^2 + 1}$$

$$r_2 = -\Delta t i - \sqrt{-\Delta t^2 + 1}$$



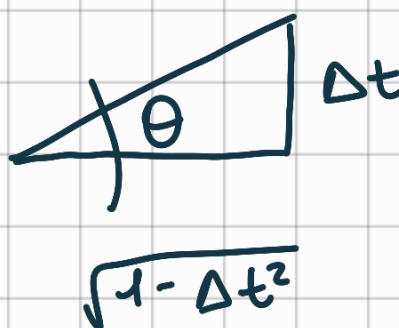
$$\Delta t < 1$$

$$|r_1| = 1 - \Delta t^2 + \Delta t^2 = 1$$

$$|r_2| = 1 - \Delta t^2 + \Delta t^2 = 1$$

$$\Delta t > 1 \quad r_1 = (-\Delta t + \sqrt{\Delta t^2 - 1})i$$

$$\Delta t > 1 \quad r_2 = -(\Delta t + \sqrt{\Delta t^2 - 1})i$$



$$\tan \theta = \frac{\Delta t}{\sqrt{1 - \Delta t^2}}$$

$$r_1 = (\sqrt{1 - \Delta t^2} - i\Delta t) = \rho e^{-i\theta} = e^{-i\theta}$$

$$r_2 = -(\sqrt{1 - \Delta t^2} + i\Delta t) = -\rho e^{i\theta} = -e^{+i\theta}$$

$$z_n = C_1 e^{-in\theta} + C_2 (-1) e^{in\theta}$$

$$z_0 = C_1 + C_2 = 1$$

$$z_1 = C_1 e^{-i\theta} - C_2 e^{i\theta} = z_0 + \Delta t F(z_0)$$

$$= 1 - \Delta t \cdot i$$

$$\begin{pmatrix} 1 & 1 \\ e^{-i\theta} & -e^{i\theta} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 - \Delta t i \end{pmatrix}$$

$$C_1 = 1 - \frac{\Delta t^2}{4} + O(\Delta t^4)$$

$$C_2 = -\frac{\Delta t^2}{4} + O(\Delta t^4)$$

$$z_n = \left(1 - \underbrace{\frac{\Delta t^2}{4}}^{\text{error}}\right) e^{-in\theta} - \underbrace{\frac{\Delta t^2}{4}}^{\text{error}} (-1)^n e^{in\theta} + O\Delta t^4$$

$$x(t_n) = \cos(t_n) = \cos(n \Delta t)$$

* Se hace filtrado (este puede ser usado para cualquier filtrado)

↳ recomendable para usar en cualquier fenómeno físico con alteración por ruido.

$$\begin{aligned} \tilde{z}_n &= \frac{1}{4} (z_{n+1} + 2z_n + z_{n-1}) \\ &= \frac{1}{4} \left(1 - \frac{\Delta t^2}{4} \right) \left[e^{-i(n+1)\theta} + 2e^{-in\theta} + e^{-i(n-1)\theta} \right] \end{aligned}$$

=

$$= -\frac{1}{4} \cdot \frac{\Delta t^2}{4} \left[(-1)^{n+1} e^{i(n+1)\theta} + 2(-1)^n e^{in\theta} + (-1)^{n-1} e^{i(n-1)\theta} \right]$$

$$\tilde{z}_n = \frac{1}{4} \left(1 - \frac{\Delta t^2}{4} \right) e^{-in\theta} \left[e^{-i\theta} + 2 + e^{i\theta} \right] - \frac{\Delta t^2}{16} (-1)^n e^{in\theta} \left[-e^{i\theta} + 2 - e^{-i\theta} \right]$$

$$\tilde{z}_n = \frac{1}{4} \left(1 - \frac{\Delta t^2}{4} \right) e^{-in\theta} [\cos\theta + 1] 2 - \frac{\Delta t^2}{16} (-1)^n e^{in\theta} [-2\cos\theta + 2]$$

$$\cos(\theta/2) = \sqrt{\frac{1 + \cos\theta}{2}} \rightarrow 1 + \cos\theta = 2\cos^2(\theta/2)$$

$$\sin(\theta/2) = \sqrt{\frac{1 - \cos\theta}{2}} \rightarrow 1 - \cos\theta = 2\sin^2(\theta/2)$$

$$\tilde{z}_n = \frac{1}{4} \left(1 - \frac{\Delta t^2}{4} \right) e^{-in\theta} 4\cos^2(\theta/2) - \frac{\Delta t^2}{16} (-1)^n e^{in\theta} \uparrow 4\sin^2(\theta/2) + O(\Delta t^4)$$

$O(\Delta t^2)$ ←

* se hace un filtrado en cada paso para
si hay presencia de spuna, atenuarla.